

Anudip Foundation 

NYC Airbnb Analysis

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Introduction

- Airbnb provides travelers with a variety of accommodation options across different price points. This project aims to analyze Airbnb listings in NYC to understand pricing trends, availability, and key factors influencing rental prices.
- This project utilizes Exploratory Data Analysis (EDA) to uncover key trends in pricing and host behavior. By analyzing the dataset, we aim to provide actionable insights for both travelers and hosts.

Data Overview

The dataset contains:

- **Total Listings:** 48,895
- **Neighborhood Groups:** Manhattan, Brooklyn, Queens, Bronx, Staten Island
- **Room Type:** Entire home/apt, Private room, Shared room
- **Key Attributes:** Price, Availability, Number of Reviews, Location

Objectives

The goal of this project is to:

- Analyze room types, prices, and availability across different neighborhoods.
- Understand host behavior and listing patterns.
- Detect potential outliers in prices.
- Provide recommendations for guests and hosts based on insights.

Dataset

- The dataset contains 20,765 entries and 22 features, including:

id: Unique identifier for each listing

name: Title of the Airbnb listing

host_name: Name: Group (borough) where the listing is located

latitude/longitude: Geolocation of listings

price: Nightly rental price

room_type: Type of accommodation (e.g., entire home, private room)

reviews_per_month: Average monthly reviews for the listing

availability_365: Number of available days in the year

Steps And Workflow

1. Data Cleaning :

- **Handle missing data:** `price`, `neighborhood`, and `beds` columns had null values.
- **Fix data types Converted :** `last\_review` to a **datetime** object.
- **Remove outliers:** Listings with prices > \$1,000 were capped to avoid skewed visualizations.

2. EDA (Exploratory Data Analysis)

1. Room type distribution:

- Visualized the count of each room type using **bar plots**.
- Identified **Entire home/apt** as the most common room type.

2. Neighborhood group insights:

- Analyzed **price variations by boroughs**.
- Manhattan had the **highest average prices**.

Steps And Workflow

3. Availability trends:

- Used **heatmaps** to show correlations among `price`, `availability\_365`, `number\_of\_reviews`, and `beds`.

4. Price distribution:

- Used **histograms** to show the distribution of prices.
- Majority of the listings were priced between \$50 - \$300.

5. Host listings:

- Analyzed hosts with multiple listings using **boxplots** to identify key contributors.

6. Review behavior:

- Used **pair plots** to show relationships between number of reviews, price, and availability.

3. Data Visualization

- **Pairplot**: To see correlations among `price`, `availability`, and `number of reviews`.
- **Heatmap**: Showing correlations among numerical features.
- **Histograms and Boxplots**: To detect outliers in `price`.
- **Bar Charts**: Displaying room types and neighborhood group distributions.

Key Functions & Libraries Used

- For data analysis, we used several Python libraries:
 1. **pandas:** Data manipulation (.read_csv(), .describe(), .groupby())
 2. **numpy:** Numerical operations (.mean(), .median())
 3. **matplotlib & seaborn:** Data visualization (histplot(), .barplot())
 4. **scikit-learn:** Machine learning for price prediction

Key Findings & Insights

1. Price Trends:

- **Manhattan** has the most expensive listings, followed by Brooklyn.
- **Entire homes/apartments** cost significantly more than private or shared rooms.

2. Room Type Distribution:

- **Entire homes/apartments** are the most common, but **private rooms** offer budget-friendly options.

3. Outliers in Price:

- Few listings priced **at \$10,000+** were detected, indicating the need to filter such extreme values.

4. Availability Patterns:

- Listings with **high availability** tend to have lower prices and more reviews, likely due to better guest experience.

5. Host Behavior:

- Some hosts manage **multiple listings**, indicating a trend toward professional hosting.

Conclusion & Recommendations

- Exploratory Data Analysis (EDA) of the New York Airbnb market highlights key trends in pricing, occupancy, and neighborhood preferences. Hosts can maximize earnings by setting competitive prices and maintaining high availability, while travelers can find budget-friendly stays in Brooklyn and Queens. Airbnb can leverage data insights to improve listing recommendations and pricing strategies, making the rental market more efficient.

Future Work

- Use machine learning to predict prices based on room type and location.
- Perform sentiment analysis on reviews to better understand guest experiences.
- Create an interactive dashboard using Plotly or Tableau for live monitoring.



Thank You!